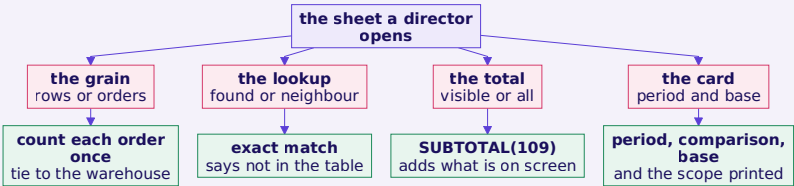


1 The sheet a director opens, and its four silent lies



Each red box prints a plausible number and no error, and the green box under it is the fix. On Friday's exports a Sum over 1,450 payment rows read Rs 39.41 crore against the warehouse's Rs 19.84 crore, C-0195 came back as a neighbour's Rs 16,740, Mumbai's foot read Rs 7,14,890 for Rs 1,56,790 on screen, and a bare Rs 19.84 crore read as a doubled quarter.

Crux. Every lie is silent, so every deliverable ships with its check, and one that fails its check is held with its reason.

2 The grain check, and the fix formula

Check	What Friday's raw export showed
Rows against ids	1,450 rows hold 1,000 orders.
Total against the warehouse	Rs 39,40,95,490 stands against Rs 19,84,00,000.
Remove Duplicates	It leaves 1,400 rows at Rs 39,40,57,740.

```
flag: =IF(COUNTIF($A$2:A2,A2)=1,1,0)
tree: =SUMIFS(order_amount, segment, "Retail-Plus",
               quarter, "Q2", flag, 1)
```

Counted once per order, the tree ties to the rupee: Rs 10.00 crore in Q1, Rs 9.84 crore in Q2.

Crux. A pivot is only as honest as the rows under it: say the grain, count rows against ids, tie the total to the warehouse.

3 A lookup has two exits

<pre>=XLOOKUP(id, ids, revenue, "not in the table") =IFERROR(INDEX(revenue, MATCH(id, ids, 0)), "not in the table") =VLOOKUP("C-0195", table, 5) 4th argument left out</pre>
XLOOKUP matches exactly by default and shows if_not_found for a missing id. VLOOKUP with range_lookup left out matches approximately, so C-0195 returned C-0194's Rs 16,740 at rank 15. Test every lookup with an id you know is missing. LibreOffice 24.2 returns #NAME? for XLOOKUP, so a sheet that must open anywhere uses the INDEX and MATCH line. Crux. A lookup that cannot find an id says so; an approximate match answers with a neighbour.

4 The foot of a filtered list

At the foot	Rows a filter hid	Rows hidden by hand
SUM	Added	Added
SUBTOTAL(9, r)	Left out	Added
SUBTOTAL(109, r)	Left out	Left out

Mumbai's 11 members spent Rs 1,56,790 while a SUM foot read Rs 7,14,890. =SUBTOTAL(102, r) counts the visible numbers the foot should be adding.

Crux. The foot of a filtered list adds only what the director can see: SUBTOTAL(109), never SUM.

5 The card and its parts

Q2, July to September 2026: Rs 9.84 crore, down 1.6 percent on Q1, April to June 2026 (Rs 10.00 crore).

A change is Q2 minus Q1 over Q1: Retail-Plus is down 29.4 percent, and 41.7 if divided by Q2. A bare Rs 19.84 crore reads as a doubled quarter.

Scope, a yellow input	The card says
All segments	Rs 9.84 crore, down 1.6%; 100.0% of revenue
All except Business	Rs 8.15 lakh, down 17.3%; 0.8% of revenue
Retail-Plus	Rs 4.13 lakh, down 29.4%; 0.4% of revenue

Crux. One number reaches the front page with its period, its comparison and its base.

6 The operating rule

Tool	What it owns
The warehouse	It owns the number, and every join, dedupe and cleaning step.
pandas	It owns the analyst's iteration.
Excel	It owns the last mile, and takes what-ifs as labelled inputs.

The drift check ties the sheet's Q2 total to the warehouse's at every refresh.

Crux. The warehouse owns the number, pandas owns the iteration, Excel owns the last mile, and nobody types over the source.

7 The director's what-if

Typed over Q2, five lakh reads down 14.6 percent against Finance's 29.4 until Monday's refresh wipes it, so the what-if goes into a yellow input beside the actual:

Line	Retail-Plus, Q2 on Q1	Where it comes from
Actual	-29.4%	The export
Scenario	-14.6%	A yellow input of Rs 5,00,000

Crux. Say yes to the question and no to the edit: the what-if sits beside the actual, never over it.

THE WORDS ON THIS SHEET

- Grain** One row stands for one thing, such as a customer, an order or a payment.

PivotTable It summarises a table by the fields you drag in, and changes when refreshed.
- Control total** It is the owner's total, which every sheet built from the data must match.

First-row flag A helper puts 1 on each order's first row, so a sum counts it once.
- Exact match** The lookup returns the row whose id equals the one asked for, or nothing.

Approximate match The lookup returns the nearest id not above the one asked for, silently.